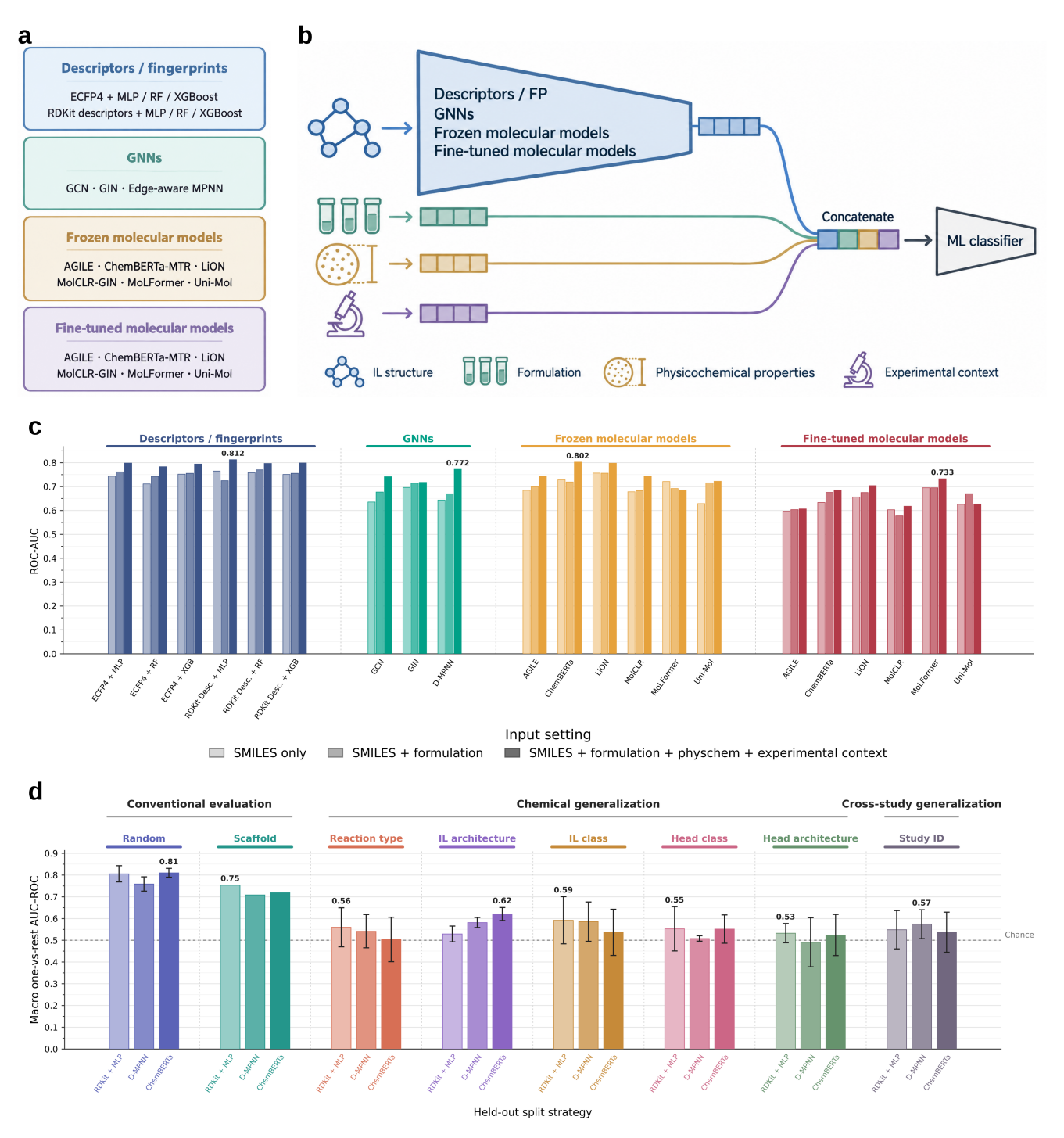

a
Descriptors / fingerprints
ECFP4 + MLP / RF / XGBoost
RDKit descriptors + MLP / RF / XGBoost
GNNs
GCN · GIN · Edge-aware MPNN
Frozen molecular models
AGILE · ChemBERTa-MTR · LiON
MolCLR-GIN · MoLFormer · Uni-Mol
Fine-tuned molecular models
AGILE · ChemBERTa-MTR · LiON
MolCLR-GIN · MoLFormer · Uni-Mol
b
Descriptors / FP
GNNs
Frozen molecular models
Fine-tuned molecular models
Concatenate
ML classifier
IL structure
Formulation
Physicochemical properties
Experimental context
c
Descriptors / fingerprints
GNNs
Frozen molecular models
Fine-tuned molecular models
ROC-AUC
0.812
0.772
0.802
0.733
ECFP4 + MLP
ECFP4 + RF
ECFP4 + XGB
RDKit Desc. + MLP
RDKit Desc. + RF
RDKit Desc. + XGB
GCN
GIN
D-MPNN
AGILE
ChemBERTa
LiON
MolCLR
MolFormer
Uni-Mol
Input setting
SMILES only
SMILES + formulation
SMILES + formulation + physchem + experimental context
d
Conventional evaluation
Chemical generalization
Cross-study generalization
Random
Scaffold
Reaction type
IL architecture
IL class
Head class
Head architecture
Study ID
Macro one-vs-rest AUC-ROC
0.81
0.75
0.56
0.62
0.59
0.55
0.53
0.57
Chance
RDKit + MLP
D-MPNN
ChemBERTa
Held-out split strategy